Symplectic Matrices in Control

Amp

A symplectic matrix preserves a bilinear form on phase space, and that single algebraic condition turns out to rigidify a spectrum, to explain why an undamped mechanical system cannot settle, and to sit underneath the Riccati equation that every LQR gain comes from. This note builds all of that from one running example: a unit mass on a unit spring, first in phase space and then under an LQR cost. We see what the condition $M^T J M = J$ preserves and forces; why the flow of an undamped system is symplectic while a damped one is not; how the condition rigidifies the spectrum and forbids asymptotic stability; how the Hamiltonian matrix of an LQR problem hands over the stabilizing Riccati solution through an invariant subspace; what a numerical reduction must preserve to compute that subspace honestly; and what breaks when a pair of eigenvalues reaches the imaginary axis.

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1. What does $M^T J M = J$ preserve?

Start in the plane, where everything can be drawn. Write a phase-space vector as $x = \binom{q}{p}$: position on top, momentum underneath. For two such vectors u and v define

$$\omega(u, v) = u^T J v, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{1}$$

Work out what this number is for $u = \binom{u_q}{u_p}$ and $v = \binom{v_q}{v_p}$:

$$u^T J v = \begin{pmatrix} u_q & u_p \end{pmatrix} \begin{pmatrix} v_p \\ -v_q \end{pmatrix} = u_q v_p - u_p v_q. \tag{2}$$

That is exactly the determinant of the 2×2 matrix with columns u and v: the **signed area** of the parallelogram the two vectors span. So in the plane, the bilinear form has a concrete meaning you can see – it measures signed parallelogram area. A matrix M is **symplectic** when

$$M^T J M = J, \tag{3}$$

and in the plane this says one visible thing: *the area of every parallelogram comes out of M unchanged.* Not just the area of one particular parallelogram – every one, because the identity holds for all pairs u, v at once.

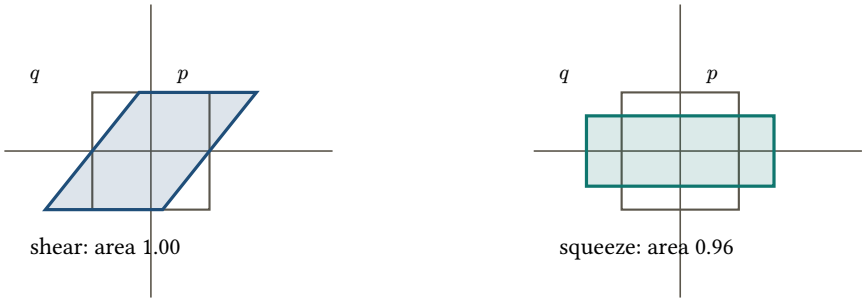


Figure 1: The unit square (gray) sent through two area-preserving-looking maps. The shear (navy) is symplectic: every parallelogram area, including the square's, survives. The squeeze (teal) multiplies all areas by 0.96, and its form defect $\|M^T J M - J\| = 0.04$ detects exactly that.

The figure contrasts two ways of being area-preserving-in-spirit. The shear $S = \begin{pmatrix} 1 & 0.8 \\ 0 & 1 \end{pmatrix}$ slides the top edge sideways; the square becomes a slanted parallelogram of the same area, and indeed $S^T J S = J$ exactly. The squeeze $D = \begin{pmatrix} 1.6 & 0 \\ 0 & 0.6 \end{pmatrix}$ stretches one axis and shrinks the other; its determinant is $1.6 \times 0.6 = 0.96$, so areas shrink by 4%, and correspondingly $\|D^T J D - J\| = 0.04$: the form defect is the area error itself. The rotation by 30° also passes, as any rotation must – it preserves all dot products, hence all areas.

Now let the algebra hand back what the condition forces. First, M must be invertible, because $\det(M^T J M) = \det M^2 \cdot \det J = \det J$ gives $\det(M)^2 = 1$. Second, the inverse has

a remarkably explicit form. Multiply Equation 3 on the left by $J^{\{-1\}} = -J$ and on the right by $M^{\{-1\}}$:

$$J = M^T J M \quad \Longrightarrow \quad -JM^{\{-1\}} = M^T J \quad \Longrightarrow \quad M^{\{-1\}} = -JM^T J. \tag{4}$$

So the inverse costs no elimination at all – a transpose and two multiplications by fixed matrices. That is the first sign that the condition is rigid, not merely restrictive. Third, $\det(M)^2 = 1$ has two roots, but the negative one never occurs for a symplectic matrix: the set of symplectic matrices is connected (it is the image of a connected group under a continuous map), and det is continuous and equals 1 at the identity, so $\det M = +1$ always. In the plane, where $M^T J M = (\det M) J$ holds for **every** 2×2 matrix M (by Equation 2 applied columnwise), symplectic and determinant-one coincide.

In $2n$ dimensions the two notions separate, and the separation is the point. A matrix with $\det = 1$ preserves the volume of $2n$-dimensional boxes. A symplectic matrix preserves something finer: the signed area of the parallelogram spanned by u and v **projected onto each of the n coordinate planes** $(q_1, p_1), \ldots, (q_n, p_n)$ – all simultaneously. The example of the 4×4 matrix that doubles q_1, halves p_1, and leaves (q_2, p_2) alone has determinant 1 but sends the (q_1, p_1)-area of a parallelogram to four times itself, so it is not symplectic. Volume preservation is one number's worth of constraint; Equation 3 is a full matrix identity, $n(2n-1)$ independent scalar equations' worth.

> **Try it:** The map $M = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ doubles all q-directions and leaves p alone. Is it symplectic? Compute $M^T J M$.

It is not: $M^T J M = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix} = 2J$, so parallelogram areas double. Compare the shear, which moves the top edge without stretching anything, and passes. Stretching one coordinate is never symplectic; the area has to be paid for elsewhere **in the same coordinate plane**.

2. Why the undamped flow is symplectic and the damped flow is not

The running example for the rest of the note is a unit mass on a unit spring. Its energy and equations are

$$H(q,p) = \frac{q^2+p^2}{2}, \quad \dot{x} = J\nabla H(x) = \begin{pmatrix} \cdot\, q \\ \cdot\, p \end{pmatrix} = \begin{pmatrix} p \\ -q \end{pmatrix}. \tag{5}$$

Read H first, before any matrices. Its two terms are the familiar energies of a spring-mass system: $\frac{q^2}{2}$ is the potential energy stored in the spring ($k=1$), $\frac{p^2}{2}$ is the kinetic energy ($m=1$). Geometrically, H is a bowl over the phase plane, and the state $x = \binom{q}{p}$ sits at height $H(x)$ on that bowl. Conservation of energy says the state never changes height: it slides **along** a level curve of H – for this H, a circle.

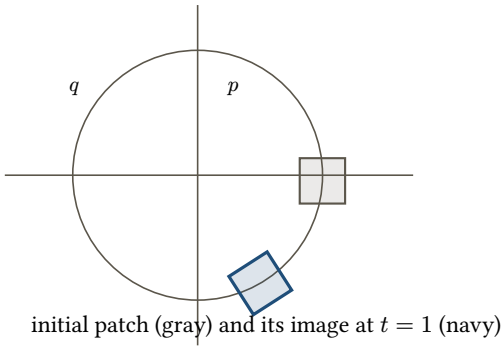
Where does the matrix form on the right come from? From Newton's laws, written as one vector equation. Velocity: $\cdot\, q = p$ (just the definition of momentum). Force: the spring pulls with minus the slope of the potential, $\cdot\, p = -\partial H/\partial q = -q$. Stacked, the pair is $\binom{p}{-q}$ – and notice the odd sign pattern: $\cdot\, q$ takes the gradient's **second** component, $\cdot\, p$ takes minus its **first**. That is a quarter-turn rotation of the gradient (linearly, "gradient" here because H is quadratic, so $\nabla H = x$ exactly). The matrix that rotates a vector a quarter turn is $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, and so the dynamics reads $\dot{x} = J\nabla H(x)$. The J is not new physics – it is the sign-differing bookkeeping that turns "energy tells the state where to go" into one equation. The format generalizes: every frictionless mechanical system, however complicated its energy, has this same shape, one antisymmetric factor J times one gradient. That is the feature the whole section leans on.

A quick consistency check from the energy itself. The rate of change of energy along the motion is $\cdot\, H = \nabla H \cdot \dot{x} = \nabla H \cdot J\nabla H$, and a dot product of a vector with its own quarter-turn is zero – a vector is perpendicular to its rotation. So $\cdot\, H = 0$: the flow moves along level curves, as promised. But energy conservation alone is too weak a property to build on: it constrains only the height of the motion, not how the motion treats a whole **neighborhood** of states. The next question is exactly about that neighborhood.

To have something concrete to test, write down the flow map Φ_t – the map sending the state at time 0 to the state at time t. Because H is quadratic here the system is linear, $\dot{x} = Jx$, and the flow map is the matrix exponential $\Phi_t = e^{Jt}$:

$$\Phi_t = e^{Jt} = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}. \tag{6}$$

A pure rotation by angle t – which is exactly what the level-circle picture predicted: a frictionless spring's state circles its energy level forever. For a general energy the flow is not a rotation (a pendulum's flow shears patches along its separatrix), which is why the argument below must not rely on rotation – only on the $J\nabla H$ structure.



initial patch (gray) and its image at $t = 1$ (navy)

Figure 2: One small square patch of the phase plane at $t = 0$ (gray, near $q = 1.1$, $p = -0.05$) and its image under the flow at $t = 1$ (navy). The patch rides a rotation by one radian around the energy circle (light gray) and lands as a rotated square of the same area – which Equation 6 guarantees, since a rotation preserves every area.

Why must this be so for **any** Hamiltonian system, not just this one? The claim is that $\Phi_t^T J \Phi_t = J$ for all t. For a linear system the flow is $\Phi_t = e^{At}$, so differentiate $F(t) = \Phi_t^T J \Phi_t$ using $\dot{\Phi}_t = A\Phi_t$:

$$\dot{F} = (A\Phi_t)^T J \Phi_t + \Phi_t^T J A \Phi_t = \Phi_t^T (A^T J + JA) \Phi_t. \tag{7}$$

Everything now hinges on the bracket. For a Hamiltonian system $A = JS$ with $S = \nabla^2 H$ *symmetric* (the energy's second derivative always is). Then

$$A^T J + JA = (JS)^T J + J(JS) = S^T J^T J + J^2 S = S + (-S) = 0. \tag{8}$$

Here is why each piece moves the way it does. $(JS)^T = S^T J^T = -SJ$: the symmetry of S lets it pass through the transpose, and $J^T = -J$ flips the sign. Then $-SJJ = -S(-I) = S$. Meanwhile $J(JS) = J^2 S = -S$. So the two terms are S and $-S$: they cancel. So $\dot{F} = 0$, and since $F(0) = J$, the form J is carried intact by the flow at every time. The structural feature responsible is precisely the pairing in $\dot{x} = J\nabla H$: one antisymmetric factor J, one symmetric factor $\nabla^2 H$, arranged so that the form's time derivative is antisymmetric-in-motion versus symmetric-in-curvature and the two kill each other.

Dissipation breaks the pairing – and it breaks the energy story at the same time. Add viscous damping $c > 0$:

$$\dot{q} = p, \quad \dot{p} = -q - cp, \quad A_c = \begin{pmatrix} 0 & 1 \\ -1 & -c \end{pmatrix}. \tag{9}$$

Run the same test. The bracket Equation 8 needs $A_c = JS$ for a symmetric S; no such S exists here (the $-c$ entry blocks it – try to solve $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} a & b \\ b & d \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & -c \end{pmatrix}$ for symmetric $\begin{pmatrix} a & b \\ b & d \end{pmatrix}$: it forces $-b = -1$ and $-b = -c$ at once), so compute the bracket directly: $A_c^T J = \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix}$ and $JA_c = \begin{pmatrix} -1 & -c \\ 0 & -1 \end{pmatrix}$, so

$$A_c^T J + JA_c = \begin{pmatrix} 0 & -c \\ c & 0 \end{pmatrix} = cJ, \tag{10}$$

which is zero only when $c = 0$. The leftover is not of the canceling form: the healthy pattern is "antisymmetric matrix times symmetric matrix," and damping's contribution resists being written that way. The energy tells the same story from its side: along the damped motion $\cdot H = \nabla H \cdot J\nabla H - cp^2 = -cp^2 \leq 0$, so the state spirals down the energy bowl instead of circling it. The two leaks are one fact seen twice – friction. The visible consequence: areas contract geometrically, at the rate given by Liouville's formula $\det e^{A_c t} = e^{\operatorname{tr} A_c \cdot t} = e^{-ct}$ – at $c = 0.5$ the area of any patch has shrunk to $e^{-0.5} = 0.6065$ after one unit of time. The damped system still converges to rest; the undamped one never does, and the next section shows that this is not an accident of this example but a theorem.

Open `Control/symplectic-matrix/symplectic-matrix-demo.html` beside this note and move the stiffness, damping, and integrator controls to watch the same patch and its area ratio.

> **Try it:** For the damped system with $c = 0.5$, is the flow Φ_t symplectic for some small $t > 0$? Use the bracket $A_c^T J + J A_c$.

No. If Φ_t were symplectic for one t_0, differentiating Equation 7 backward would still force the bracket to vanish at $t = 0$ (the derivative $\dot{F}(0) = A_c^T J + J A_c$ is independent of t), giving $\begin{pmatrix} 0 & -c \\ c & 0 \end{pmatrix} = 0$, a contradiction for $c > 0$. The flow is symplectic at no positive time whatsoever – symmetry is lost instantly, not gradually.

3. How the condition rigidifies the spectrum

The inverse formula Equation 4 is the lever. Read it as a similarity:

$$M^{\{-1\}} = -JM^TJ = JM^TJ^{\{-1\}}, \tag{11}$$

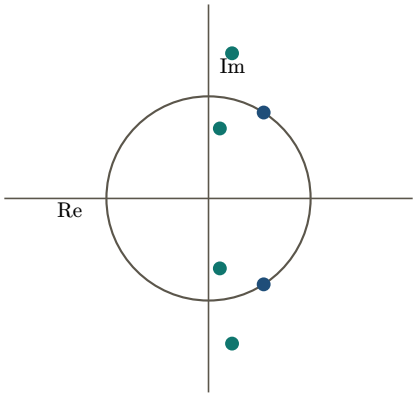
$M^{\{-1\}}$ is similar to M^T. And M is always similar to M^T (a matrix and its transpose share the characteristic polynomial). Chaining the equalities:

$$\sigma\big(M^{\{-1\}}\big) = \sigma\big(M^T\big) = \sigma(M). \tag{12}$$

The spectrum of M equals the spectrum of its inverse. If λ is an eigenvalue, so is $\frac{1}{\lambda}$. And because a symplectic matrix is real, $\overline{\lambda}$ is an eigenvalue too. Putting the two facts together, eigenvalues come in **quartets**:

$$\lambda, \quad \frac{1}{\lambda}, \quad \overline{\lambda}, \quad \frac{1}{\overline{\lambda}}. \tag{13}$$

Whenever one eigenvalue sits somewhere in the complex plane, three more occupy fixed positions in the same matrix's spectrum: the reciprocal point across the real axis, the mirror across the unit circle, and both combined. The figure shows two unrelated symplectic matrices, and in each spectrum the quartet structure is plainly visible.



unit circle; navy M_1, teal M_2

Figure 3: Eigenvalues of two symplectic matrices: $M_1 = \Phi_1$ of Equation 5 (navy, both on the unit circle) and $M_2 = e^{H_2}$ for a 4×4 Hamiltonian H_2 (teal). For M_2, the two visible points $0.1111 + 0.6853i$ and $0.2306 + 1.4218i$ are reciprocals of each other: $0.1111 + 0.6853i = \frac{1}{0.2306+1.4218i}$ (product $0.2306 \times 0.1111 + 1.4218 \times 0.6853 = 1.0000$). Each teal point also has its conjugate partner, drawn.

For a discrete system $x_{k+1} = Mx_k$, asymptotic stability means every eigenvalue inside the unit circle. The quartet rule forbids it outright: if $|\lambda| < 1$ then $|\frac{1}{\lambda}| > 1$, so an eigenvalue strictly inside the circle drags a partner strictly outside. The best a symplectic matrix can do is keep its whole spectrum **on** the unit circle (as M_1 does), or split it half-in half-out (as M_2 does). Translated back to mechanics: an undamped Hamiltonian system cannot have all its modes decay. This is the spectrum-level statement of "frictionless systems do not settle," and it holds no matter how the energy is shaped, because only Equation 3 was used.

> **Try it:** An eigenvalue of a symplectic matrix is reported as $0.5 + 0.5i$. Which other numbers must be in the spectrum, and why can the matrix not be asymptotically stable?

The quartet Equation 13 also contains $0.5 - 0.5i$ (conjugate), $\frac{1}{0.5+0.5i} = 1 - i$, and $1 + i$ – magnitudes 0.7071 and 1.4142, one inside the circle and one outside. No symplectic matrix is asymptotically stable: the reciprocal partner always escapes.

4. The Hamiltonian matrix of an LQR problem and its stable subspace

Now put the same oscillator under a cost. The LQR problem: choose u to minimize

$$\int_0^\infty (x^T Q x + u^2)\,\mathrm{d}t, \quad \dot{x} = Ax + Bu, \tag{14}$$

with the running-example data $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)$, $Q = \varepsilon I$, $R = 1$. The classical result – used here, not re-derived – is that the optimal control is $u = -Kx$ with $K = B^T P$ for the **stabilizing** symmetric solution P of the algebraic Riccati equation

$$A^T P + PA - PGP + Q = 0, \quad G = BR^{\{-1\}}B^T = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \tag{15}$$

The equation is quadratic in P, nonlinear, and nothing in its statement hints at the symplectic structures of the first three sections. The bridge is a $2n \times 2n$ matrix built from the problem data:

$$H = \begin{pmatrix} A & -G \\ -Q & -A^T \end{pmatrix}. \tag{16}$$

H is **Hamiltonian**, meaning JH is symmetric (with the $2n \times 2n$ form $J = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$):

$$JH = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}\begin{pmatrix} A & -G \\ -Q & -A^T \end{pmatrix} = \begin{pmatrix} -Q & -A^T \\ -A & G \end{pmatrix}, \tag{17}$$

which is symmetric, because Q and G are. So $H = -J(JH)$ (using $J^2 = -I$) is exactly the "antisymmetric times symmetric" pattern from Section 2, at the matrix level: the same pairing that made flows symplectic.

The Hamiltonian identity has an immediate spectral consequence, proved the same way as Equation 12 but with a minus sign. For Hamiltonian H, one checks $H^T = JHJ^{\{-1\}} \cdot (-1)$ – precisely, $-H = JH^T J^{\{-1\}}$ – so H is similar to $-H$, and

$$\sigma(H) = -\sigma(H): \quad \lambda \in \sigma(H) \implies -\lambda \in \sigma(H). \tag{18}$$

With real H, the full pattern is the quartet $\lambda, \overline{\lambda}, -\lambda, -\overline{\lambda}$: reflection through the origin has joined the picture. Where the symplectic **matrix** of Section 3 paired points across the unit circle, the Hamiltonian **matrix** pairs them across the imaginary axis. If no eigenvalue of H lies on that axis, the $2n$ eigenvalues split evenly: exactly n in the open left half-plane, n in the open right half-plane. For the running example at $\varepsilon = 1$ the characteristic polynomial is $x^4 + x^2 + 2$, with roots $\pm 0.6761 \pm 0.9783i$ – two left, two right, as promised.

Now the step that turns this into a Riccati solution. Take the n left-half-plane eigenvectors of H; they span an n-dimensional **invariant subspace** – a subspace that H maps into itself. Stack a basis of it as two $n \times n$ blocks:

$$H\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}\Lambda, \quad \text{Re eig}(\Lambda) < 0. \tag{19}$$

Multiply the block rows out, using Equation 16:

$$AX_1 - GX_2 = X_1\Lambda \quad \text{and} \quad -QX_1 - A^T X_2 = X_2\Lambda. \tag{20}$$

Assume for now X_1 invertible (this is the generic case; Section 6 shows exactly when it fails) and set $P = X_2 X_1^{\{-1\}}$. Three things happen at once.

Stability. Multiply the first block equation by $X_1^{\{-1\}}$ on the right:

$$A - GP = X_1 \Lambda X_1^{\{-1\}}. \tag{21}$$

The closed-loop matrix $A - GP$ is similar to Λ, hence has exactly the stable spectrum. The candidate P delivers a stabilizing controller.

Riccati. The second block equation, with $X_2 = PX_1$ and $X_2\Lambda = PX_1\Lambda = P(A - GP)X_1$ by Equation 21, reads $-QX_1 - A^T PX_1 = P(A - GP)X_1$. Cancel X_1:

$$A^T P + PA - PGP + Q = 0. \tag{22}$$

P satisfies the Riccati equation – the cancellation of X_1 is the whole trick, and it is why P is defined as $X_2 X_1^{\{-1\}}$ in the first place.

Symmetry. The Riccati solution must be symmetric; here the symplectic structure finally cashes in. Take two eigenvectors $Hu = \lambda u$ and $Hv = \mu v$ from the stable subspace. Form $u^T JHv$ two ways, using first $Hv = \mu v$ and then the symmetry of JH (Section 2's identity, now for the matrix H):

$$u^T JHv = \mu \cdot u^T Jv = (JHu)^T v = \lambda \cdot u^T J^T v = -\lambda \cdot u^T Jv. \tag{23}$$

Comparing ends: $(\mu + \lambda) \cdot u^T Jv = 0$. For two stable eigenvalues, $\lambda + \mu$ has negative real part, hence is not zero – so $u^T Jv = 0$ for every pair of stable eigenvectors. Applying this to the columns of $\begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$:

$$X^T JX = 0 \quad \text{with} \quad X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \quad \Longleftrightarrow \quad X_1^T X_2 = X_2^T X_1. \tag{24}$$

(A subspace whose basis satisfies $X^T JX = 0$ is called **Lagrangian**; the identity above is its algebra.) Then $P = X_2 X_1^{\{-1\}}$ has transpose $P^T = X_1^{\{-T\}} X_2^T = X_1^{\{-T\}} X_1^T X_2 X_1^{\{-1\}} = X_2 X_1^{\{-1\}} = P$. Symmetric, solving the equation, stabilizing: the construction delivers exactly the matrix the LQR theorem asks for. For the running example at $\varepsilon = 1$ it produces $P = \begin{pmatrix} 1.9123 & 0.4142 \\ 0.4142 & 1.3522 \end{pmatrix}$ and the closed loop $A - GP$ with eigenvalues $-0.6761 \pm 0.9783i$ – the left half of $\sigma(H)$, as Equation 21 predicted.

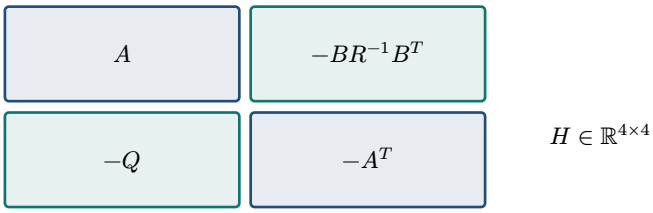


Figure 4: The block layout of Equation 16 for the running example, each block labeled by the problem data it carries. The diagonal blocks A and $-A^T$ mirror each other; the off-diagonal blocks $-G$ and $-Q$ are symmetric – this is the matrix-level image of "antisymmetric times symmetric."

Try it: Suppose the stable invariant subspace had been spanned by $\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$ (i.e. $X_1 = I_2$ stacked above $X_2 = I_2$). What controller does $P = X_2 X_1^{\{-1\}}$ give, and what is wrong with it?

$P = I_2$, so $K = B^T P = \begin{pmatrix} 0 & 1 \end{pmatrix}$ and the closed loop is $A - GP = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$, which is stable – but P would also have to satisfy Equation 15, and the subspace here is not an invariant subspace of the true H: its columns are not eigenvector combinations for the stable pair $-0.6761 \pm 0.9783i$. The construction is only as good as the subspace: an arbitrary stable-looking X_1, X_2 satisfies neither the Riccati equation nor, generally, the Lagrangian identity Equation 24.

5. What a numerical reduction must preserve

The construction of Section 4 needs one thing from linear algebra: the stable invariant subspace of H. The standard route is a Schur decomposition – an orthogonal change of basis Q making $Q^*HQ = T$ upper triangular. But an ordinary Schur decomposition is the wrong instrument here, for a reason worth seeing precisely. Two demands collide.

The first demand is *separation*. The Schur form is triangular, and its diagonal carries the eigenvalues in whatever order the algorithm happens to produce them. The stable subspace is spanned by the leading columns of Q **only if** the first n diagonal entries of T are exactly the stable eigenvalues. Nothing in the ordinary Schur iteration promises that order: it may interleave stable and unstable eigenvalues along the diagonal, in which case the "leading block" is a mixture and the P read off from it is meaningless.

The second demand is *structure*. The Riccati solution's symmetry came from the Lagrangian identity Equation 24, which came from the form J. An orthogonal Q preserves the Euclidean form – $Q^*Q = I$ – but knows nothing about J; after reduction, the relation between the $(1,2)$ and $(2,1)$ blocks of H that made everything work is no longer visible, and roundoff can drift it arbitrarily far. The computed P then fails to be exactly symmetric, not by a rounding-scale error in the entries but by loss of the property the equation is about.

A Hamiltonian Schur decomposition fixes both at once by restricting the allowed transformations:

Reduction	Transformed form	Basis and what it is built from
ordinary Schur	$Q^*HQ = T$, Q unitary, T triangular	columns of Q; stable and unstable eigenvalues may be interleaved along the diagonal
Hamiltonian Schur	$U^THU = \begin{pmatrix} T_{11} & T_{12} \\ 0 & -T_{11}^T \end{pmatrix}$, U symplectic-orthogonal	first n columns of U; the bottom-left block is zero and $T_{22} = -T_{11}^T$ links the diagonal blocks through the form

The transformation U is required to satisfy $U^TJU = J$ – Section 1's condition, applied to the change of basis itself. This is the same move as throughout the note: the form is the invariant, so only form-preserving maps may be used. Under such a U, the transformed H remains Hamiltonian, and the Hamiltonian Schur **form** above makes the structure explicit: the $(2,1)$ block is zero and the $(2,2)$ block is $-T_{11}^T$, the mirror of the $(1,1)$ block – Equation 18 written into the shape of the matrix. The eigenvalue order is no longer incidental: the leading block carries exactly one member of each $\pm$ pair, and algorithms (orthogonal symplectic Hessenberg–Schur iterations, following Laub's method and its descendants) arrange for it to be the stable one. Then the first n columns of U, stacked as $\begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$, span the stable invariant subspace with the Lagrangian identity holding up to rounding, and $P = X_2X_1^{\{-1\}}$ comes out symmetric and well-conditioned whenever the problem itself is. The step that reads P off

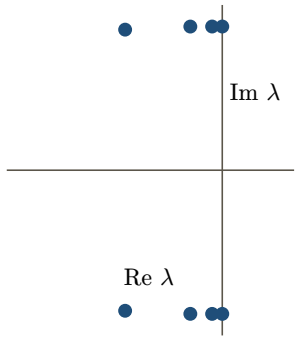
needs X_1 invertible and not nearly singular – a demand the ordinary Schur route can violate silently even when P is perfectly fine.

> **Try it:** An ordinary Schur decomposition of the running-example H returns T whose first two diagonal entries are $+0.6761 \pm 0.9783i$ (the right-half-plane pair). What does $P = X_2 X_1^{\{-1\}}$ from the leading columns compute?

It computes the **antistabilizing** solution: the invariant subspace now belongs to the unstable eigenvalues, so the resulting P solves the Riccati equation but makes $A - GP$ have the unstable spectrum $\pm 0.6761 \pm 0.9783i$. The equation has several symmetric solutions; only ordering the reduction correctly selects the controller. This is why eigenvalue ordering, not just accuracy, is part of the numerical contract.

6. When an eigenvalue lands on the imaginary axis

Every step so far had one hypothesis quietly doing load-bearing work: H has no eigenvalues on the imaginary axis. It bought the clean split of Equation 18 into left/right halves, and it is exactly what makes the stable invariant subspace – and with it P – exist. So watch what happens as the state penalty shrinks. Keep Equation 14 and Equation 16 but let $Q = \varepsilon I$ with $\varepsilon \to 0$. The table and figure track the left-half-plane member of the closed-loop pair; the full spectrum of H is that pair together with its conjugates and their negations.



ε	λ OF $\sigma(H)$ IN THE LHP	$\lvert\lambda\rvert$
1	$-0.6761 + 0.9783i$	1.1892
0.1	$-0.2223 + 0.9997i$	1.0241
0.01	$-0.0707 + 1.0000i$	1.0025
0	$0 + 1.0000i$	1.0000

Table 2: The left-half-plane eigenvalues of H as ε decreases: from $-0.6761 \pm 0.9783i$ at $\varepsilon = 1$ the pair slides toward the imaginary axis and lands on $\pm i$ at $\varepsilon = 0$. The real part, which is what separates stable from unstable, goes to zero.

The mechanism behind the migration: at $\varepsilon = 0$ the Riccati equation Equation 15 becomes $A^T P + PA - PGP = 0$, and $P = 0$ solves it. The closed loop is then A itself, with eigenvalues $\pm i$ – the undamped oscillator of Section 2, which we already knew cannot be moved by any feedback that costs nothing to leave alone. On the Hamiltonian side, the four eigenvalues of Equation 18 collide in pairs on the imaginary axis: the left-half-plane member and its right-half-plane mirror meet at the same point, and the split into "n stable, n unstable" loses its meaning – there are no left and right halves to count into.

The subspace construction then fails at its stated assumption. With the stable and unstable eigenvalues merged, there is no n-dimensional invariant subspace of stable eigenvectors to span, no Λ with spectrum in the open left half-plane, and X_1 loses its invertibility – indeed for $\varepsilon = 0$ the "solution" collapses to $P = X_2 X_1^{\{-1\}} = 0$ with the closed loop marginally stable, not asymptotically. The general statement: a stabilizing solution of the Riccati equation exists precisely when H has no imaginary-axis eigenvalues; its position in the complex plane is not a numerical curiosity but the exact boundary between "the controller exists" and "it does not." This is also why practical LQR design never uses $Q = 0$: an unweighted state returns the marginal system, and any reduction method (Section 5) would see colliding eigenvalues where the splitting it relies on has dissolved.

> **Try it:** At $\varepsilon = 0.01$, the LHP eigenvalue is $-0.0707 + 1.0000i$. Roughly how far, in real part, does the pair sit from the imaginary axis, and what does the Hamiltonian Schur reduction face as ε shrinks further?

The real part is -0.0707, and it scales roughly linearly with ε (compare -0.2223 at $\varepsilon = 0.1$). The reduction must separate eigenvalues whose real parts differ by $O(\varepsilon)$: as the collision approaches, the invariant subspace becomes increasingly sensitive, and the computed X_1 grows toward singularity – the well-conditioned regime of Section 5 shrinks with ε.

7. Sources and further reading

The symplectic-group and Hamiltonian-flow material is classical (Arnold, **Mathematical Methods of Classical Mechanics**); the invariant-subspace construction of the Riccati solution follows Laub, **A Schur method for solving algebraic Riccati equations** (1979), and the structure-preserving reduction of Section 5 descends from Laub's method via Byers, **A Hamiltonian QR algorithm** (1983), and the symplectic-method literature of Benner, Mehrmann, and coauthors. Every number above is arithmetic on the running example, reproduced by `Control/symplectic-matrix/scripts/verify_symplectic.py`.